

Atul Pal

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PROFILE

Software Engineer and 2026 B.Tech graduate experienced in backend development, distributed systems, cloud, and AI. Improved API latency by 32%, database read performance by 40%, and built a Redis service benchmarked at 7K+ req/s. Solved 400+ DSA problems on LeetCode [[Profile Link](#)].

EDUCATION

Netaji Subhas University of Technology (Formerly NSIT) <i>Bachelor of Technology in Electronics and Communication Engineering</i>	New Delhi, India Aug 2022 – May 2026
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EXPERIENCE

Research Intern <i>Netaji Subhas University of Technology (NSUT)</i>	May 2025 – Jul 2025 New Delhi, India
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- Researched deep-learning-based skin lesion classification on the HAM10000 dataset (10K+ images), benchmarking custom **CNN, ResNet50, and EfficientNetB0** architectures.
- Improved classification performance to **93.6% accuracy and 0.92 F1 Score** using EfficientNetB0 transfer learning, compared with **89.1%** for ResNet50 and **82.4%** for the baseline CNN
- Built model evaluation and explainability workflows using **precision, recall, F1 score, and confusion matrix, and Grad-CAM** to analyze predictions and visualize influential lesion regions.

Software Engineer Intern <i>Square Yards</i>	May 2024 – Jul 2024 Gurugram, India
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- Developed and optimized backend REST APIs using Node.js and Express.js for property listings and user workflows, reducing average API response latency by **32% (280 ms to 190 ms)**
- Optimized MongoDB queries and indexing for production property-listing APIs, reducing database read latency by **40%** and improving response performance under concurrent traffic.
- Delivered **8+ production features in 3 months** through sprint planning, code reviews, daily standups, and cross-functional Agile collaboration.

PROJECTS

Distributed Rate Limiter Service <i>Node.js, Express.js, Redis, Docker</i>	Apr 2026 – Jul 2026
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- Designed and implemented a distributed rate limiting service using **Redis, Lua scripting, and Token Bucket**, ensuring **atomic operations** and eliminating race conditions under high concurrency.
- Architected the service across **stateless Node.js instances** with centralized Redis state, applying distributed systems concepts and CAP trade-offs.
- Integrated **Prometheus, Grafana, and k6** for observability and load testing, sustaining **7K+ requests/sec** during benchmark tests.

SkinScan Live <i>Python, FastAPI, Next.js, TensorFlow, MongoDB</i>	Sep 2025 – Dec 2025
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- Built an end-to-end AI web platform using **FastAPI, Next.js, and MongoDB Atlas**, integrating model inference, authentication, RBAC, Cloudinary storage and external APIs.
- Engineered a **CNN → Grad-CAM → LLM** inference pipeline with confidence scoring and **end-to-end response time under 3 seconds**.
- Fine-tuned and served an **EfficientNetB0 model for 7 skin-lesion classes**, achieving **85% validation accuracy** while generating explainable predictions using Grad-CAM.

AWS Serverless Deployment Pipeline <i>Node.js, AWS Lambda, S3, GitHub Actions</i>	Sep 2024 – Nov 2024
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- Built a serverless CI/CD pipeline using GitHub Actions, AWS Lambda, API Gateway, S3 and IAM to automate build, test and deployment workflows.
- Automated deployments on every commit, reducing release time from **30 minutes to under 5 minutes** and manual effort by **80%**.
- Implemented IAM least-privilege policies, CloudWatch monitoring and event-driven S3/Lambda processing for secure and isolated deployment stages.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, SQL

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT, Redis, MongoDB, PostgreSQL

Cloud & DevOps: AWS, Docker, GitHub Actions, Linux, Prometheus, Grafana

Frontend: React, Next.js, HTML/CSS

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks